

Mini Case Study – Portfolio Task 1

You are a data analytics intern at a company that conducted an advertising experiment to measure the effectiveness of a digital advertisement campaign on a website. The experiment was implemented by storing a cookie on the website visitor's device before serving the first ad, that randomly determined whether the visitor belonged to the treatment or control group. The treatment group was eligible to see ads, while the control group did not see any ads. The company also stored additional information about ad exposure (impressions). For the treatment group, the company stored the number of times the ad was loaded when the user visited the website during the campaign period. For the control group, they stored the number of times the user would have been exposed to the ad if they were not in the control group, even though no ad was displayed to them (similar to Ghost ads and ensuring compliance in the control group).

The company tracked whether users from the experiment made a purchase in their online store (using a tracking cookie). The average contribution of a purchase is estimated to be 200 NOK. The cost of advertising on the website was 100 NOK for every 1000 ad impressions.

Your job is now to carefully analyze the data and provide answers to the following questions for the marketing manager:

1. Is there evidence for non-compliance in the data? If yes, what type of non-compliance and how does it affect your ability to estimate different types of treatment effects?
2. Estimate the different types of treatment effects (e.g., ATE, ATET, LATE, ITT; *depending on your previous conclusions*) of showing the ad on the conversion rate and communicate them appropriately for a managerial audience.
3. Analyze the effect of impression on the treatment effect (i.e., estimate CATEs). How does this supplement or change your previous findings?
4. Is showing ads profitable for the company?
 - Calculate the net contribution of an ad exposure (taking into account the ad effect, average exposure in the control group, and ad costs per exposure)
 - Perform some sensitivity assessments (at which average contribution, ad costs, or ad effect will showing ads be unprofitable?)
5. Critically discuss the methods you use to estimate the treatment effects and their inference statistics. What are limitations and other options?
6. Discuss potential violations of assumptions for causal inference (e.g., SUTVA) in this context and assess their potential impact on the size of the effects.

Format:

- Report the analyses including the code and some discussions about what you did and your key findings (e.g., an R-Studio based qmd report).
- Write a short conclusion at the end summarizing the key findings of your analysis and some potential suggestions for the marketing manager.

Submission:

- You have the chance to get feedback by submitting your first analysis by Feb 9 via email.
- Integrate your final report into the portfolio as one well-organized document.
- Submit the final version as part of the portfolio in Wiseflow.

Assessment dimensions

1. Use of Terminology and Concepts:
 - Uses appropriate terminology and concepts from the course or class literature.
 - Uses appropriate methods from the course or class literature.
2. Data Analysis:
 - Demonstrates understanding of the data and applies appropriate analysis techniques.
3. Communication and Organization:
 - Provides clear answers to the questions given in the task.
 - Presents code and analysis in a clear and organized manner.
 - Presents well-developed discussions about analysis findings.
4. Reflection and Suggestions for the Marketing Manager:
 - Shows awareness for the challenges in the data analysis and provides careful and practical recommendations.
 - Overall, demonstrates critical thinking, synthesis of information, and the ability to draw meaningful conclusions.

AI / LLMs:

- Generally, follow the “Guidelines for student use of AI at BI”
<https://portal.bi.no/globalassets/tjenester/ki-verktoy/guidelines-for-student-use-of-ai-at-bi-2025-2026.pdf>
- You will achieve the learning outcomes best by completing the tasks yourself through independent analysis, reasoning, and writing. Therefore, use AI tools as little as possible. Appropriate use may include limited language editing (e.g., improving clarity or grammar) or helping you find the right R syntax for analysis that you want to adopt. Inappropriate use includes generating or substantially shaping the content, such as proposing the analytical approach or analysis code, interpreting results and outcomes, developing arguments, solving problems, or producing text that you submit as your own.
- Include a short reflection on your use of AI (if applicable):
 - Description of AI use
 - Which tools you have used and for what purposes (e.g., language editing, information search, etc.).
 - How you have processed and further developed AI-generated contributions.
 - Evaluation of the learning process
 - How AI tools have supported your academic development.
 - Which choices you made in the interaction between your own knowledge and AI support.
 - Quality assurance
 - How you have assessed and verified information from AI tools.
 - Which academic judgments you made to ensure quality and relevance.

Any open questions? Please ask on Ed Discussion.